

Number : JODDB/Sales/2021/5111

Date : ٨ November 2021

United Glass Industries Company (uniglass)
Subject: Test Report

Ref: Our letter number JODDB/Sales/2021/4630 dated 4/10/2021.

Dear Sir,

1. Kindly find attached Test report no. JODDB/Test/BTF/TR/981 which represents the result for the test in the reference above.
2. We need to receive your kind feedback regarding attached test report by filling the attached Customer feedback Survey.
3. Thank you for your valued business and look forward hearing from you soon.

Sincerely Yours,



On behalf of CEO Jordan Design & Development Bureau
Head of Sales & Commercial Activities Branch
Eng. Rateb A. Abu Al Ragheb

Cc:

- Acting Head of Test and Evaluation Department
- CEO office manager / FYI

Test & Evaluation Department

Ballistic Testing Facility

Test Report

Ballistic Resistance Test

65.5 mm Ballistic Glass

**Ballistic Resistant Protective Materials “NIJ Standard 0108.01 Level III”
(7.62x51mm NATO Ball “M80”)**

United Glass Industries Company /KSA

JODDB/TEST/BTF/TR/981



4515

This report relates only to the test specimens tested and should not be published or copied or distributed or reissued to any party whatsoever without the prior written permission of Jordan Design and Development Bureau.

Approval

Test Report

65.5 mm Ballistic Glass

Ballistic Resistant Protective Materials "NIJ Standard 0108.01 Level III" (7.62x51mm NATO Ball "M80")

United Glass Industries Company /KSA

Compiled by
Head of Weapons, Ammunition & Armor Testing
Dr. Eng. Riyadh Ali Ratrout

Approved by
Acting Head of Test & Evaluation Department.
Dr. Eng. Riyadh Ali Ratrout


Signature

4 November 2021
Date

1. Test Data

Requester	JODDB/ Sales
Contractor	United Glass Industries Company ; P.O.Box 40179, Riyadh, 11499 KSA
References	• Test and Evaluation Request No. 981 dated 3/10/2021 • Test Proposal No. JODDB/TEST/BTF/TP/981 Dated 4/10/2021 (See Appendix A)
Test Sample	65.5 mm Ballistic Glass/Weight 38.75 kg/ Size 500x500mm See Photo No.1
Manufacturer	United Glass Industries Company /KSA
Test Type	Ballistic Resistance Test
Test Standard	Ballistic Resistant Protective Materials "NIJ Standard 0108.01 Level III"
Test Site	Ballistic Testing Facility (JODDB)
Sample Received Date	31 th October 2021
Test Temperature	22 °C
Test Weapon	NATO Universal Ballistic Breech with standard barrel:7.62x51mm SN:5043
Type of Bullets	7.62x51mm NATO Ball "M80", Batch No.171-GGG-14
Test Equipment	• Doppler Radar SN.3503-13-270-011 (± 0.4 m/s) • Digital calliper SN.A18229682 (± 0.05 mm) • Temp .Hum. Meter SN.95941255 (± 0.3 °C) • Distance laser Measuring Device (Lacia) SN. 1030969032 (± 2 mm)
Test Conducted By	• Eng. Issa Rawashdeh • Eng. Ali Al Sardyah • Eng. Sameer Obeidat • Senior Technician Jalal Gholilat • Technician Tariq Al. Momani • Technician Moath Anagreh
Test Date	31 th October 2021

Note: The results contained in this report are only valid for detailed above and the report shall not be reproduced except in full without approval of the laboratory.

2. Test Results

Test Sample				
65.5 mm Ballistic Glass				
NIJ 0108.01 Standard / Level III				
Threat	Shot No.	Bullet Velocity @ 2.5m (m/s)	Angle of Incidence (°)	Status (P/ NP) P- Penetration NP- No Penetration
(7.62x51mm) NATO Ball, (M80) @ 15 m Bullet Velocity "838±15 m/s"	1	843.5	0°	NP
	2	840.9	0°	NP
	3	843.3	0°	NP
	4	839.5	0°	NP
	5	845.2	0°	NP
No Penetration. See photos No.2				
Status: Comply ; See applied decision rule page 5				
Criteria: Penetration: Perforation of a witness plate by any part of the test specimen or test bullet. No Penetration: No perforation of a witness plate by any part of the test specimen or test bullet.				

RANGE EQUIPMENT & CONFIGURATION

THE GUN

The test sample was shot from the NATO Universal Ballistic Breech with the appropriate standard barrel and bullet type to give projectile stability.

VELOCITY MEASUREMENT

The projectile velocity was measured using Doppler Radar System at 2.5m from the muzzle with uncertainty 0.4 m/s

Applied Statement of Conformity and Decision Rule

Test Level	Test Result	Applied Decision Rule	Statement of Conformity
NIJ 0108.01 Standard Level III 7.62x51mm NATO Ball "M80"	No Penetration	No Perforation of a witness plate by any part of the test specimen or test bullet When Bullet velocity@ 2.5m ≥ 824.4 m/s.	Comply

DISTANCE MESUREMENTS

Test range distance was measured by laser measuring device with uncertainty ± 2 mm, striking distances were marked by a digital calibre with uncertainty ± 0.05 mm.

SUPPORT FIXTURE

The test sample was supported by a fixture that permits its position and attitude to be readily adjusted so that it is perpendicular to the line of flight of the bullet at the point of impact.

AIMING

The aiming process has been achieved by using standard laser bullets.

SHOT PLACEMENT

All shots hit the test sample in the required pattern, with the required place.

SPLINTER COLLECTING BOX & WITNESS SHEET

An aluminium sheet with a thickness of 0.5 mm (AlCuMg1, F 40) has to be used as the penetration indicator. It has to be fixed in a distance of 150 mm ± 5 mm behind the test Specimen.

Photo No. 1: Test Sample

65.5 mm Ballistic Glass

Impact Side



Isometric View

Photo No. 2: Status of Test Sample after firing 7.62x51mm "M80" rounds from 15m.

65.5mm Ballistic Glass size 500x500 mm (No Penetration)

Impact Side



Rear Side



Witness sheet



APPENDIX A: Approved Technical Testing Proposal

Contractor	United Glass Industries Company (Uniglass) / KSA			
Requester	JODDB / Sales			
Reference	Test and Evaluation Request No. 981 dated 03/10/2021			
Test Sample	Test Type	Test Reference	Test Requirements	Test Criteria
Ballistic Glass NU 0108.01 LEVEL III 500x500mm Qty. 2	Ballistic Resistant Test	Ballistic Resistant Protective Materials NU Standard 0108.01 Level III "Accredited Test"	Firing five rounds type 7.62x51mm NATO Ball "M80" on impact side of a test sample. Target Distance: 15 m. Impact Velocity : 838 ± 15 m/s (Doppler Radar) Firing Pattern "Pentagon"	See Annex A

Note: It is necessary to provide JODDB with thickness, serial number of each test sample and also mark impact side prior to testing.

Annex A

Test Criteria Statement of Conformity and Decision Rule			
Test	Comply:	Comply with conditions:	Not comply:
Level III 7.62x51mm NATO Ball "M80"	No perforation of a witness plate by any part of the test specimen or test bullet when Bullet velocity @ 2.5m ≥ 824.4 m/s.	No perforation of a witness plate by any part of the test specimen or test bullet when 823 m/s ≤ Bullet velocity @ 2.5m < 824.4 m/s	Perforation of a witness plate by any part of the test specimen or test bullet when Bullet velocity @ 2.5m ≤ 851.60 m/s.

SLS05-01

BTT Customer Feedback Survey

Issued by: Qulaity Assurance

Effective Date: 24/01/2021

Rev. 1

Pg. 1 of 1

Contractor	
Reference	
Contact Information	

According to ISO 17025:2017 requirements and in order to improve our services, Appreciate your cooperation by filling the following survey at your convenient time

QUALITY ELEMENT	Excellent	Good	Average	Fair	Poor
Technical Competence					
Turn-around Testing Time					
Communication with laboratory's employees					
Overall Quality Rating					
Compared to your experience with other laboratory providers, JODDB Testing Facilities services are					

Additional Comments:

Name:

Date:

Job Title/Function: